Eine Woche, ein Beispiel

4.26. moduli of v.b. over curve

This document is mainly a table of explicit description for moduli of v.b. over curve.

Ref:

[Beau94]: Arnaud Beauville. "Vector Bundles on Curves and Generalized Theta Functions: Recent Results and Open Problems." arXiv:alg-Geom/9404001. Preprint, arXiv, April 5, 1994. https://doi.org/10.48550/arXiv.alg-geom/9404001.

[Beau05]: Arnaud Beauville. "Vector Bundles on Curves and Theta Functions." arXiv:math/0502179. Preprint, arXiv, February 11, 2005. https://doi.org/10.48550/arXiv.math/0502179.

For a sm proj curve $C/\mathbb{C}$ with $g(C)=g$, we want to describe

$$M_C(r,\mathcal{O}_C) = \{\text{semistable v.bs on } C \text{ with rk } r\}/\sim$$

Also written as $SU_C(r)$

One can construct a rational map

$$\varphi: M_C(r,\mathcal{O}_C) \dashrightarrow |r\Theta_C| \cong \mathbb{P}^{r^g-1}$$
$$\mathcal{E} \longmapsto \Theta_{\mathcal{E}} = \{\mathcal{L}\in \mathrm{Pic}^{g-1}(C) \mid H^0(\mathcal{E}\otimes\mathcal{L}) \neq 0\}$$

Well-defineness:

① Writing $\Theta_{\mathcal{E}}$ as the determinant loci, we get $\Theta_{\mathcal{E}}$ is of codim ≤ 1;

② When $\Theta_{\mathcal{E}}$ is a divisor, use Porteous' formula to show that $[\Theta_{\mathcal{E}}] = [r\Theta_C]$;

③ verify that $\varphi(\mathcal{O}_C^{\oplus r}) = r\Theta_C$ is a divisor.

Conj. [Beau 05, Prop 4.5]
φ is generically finite.

$g \backslash r$	1	2	3	4
1	0 0	1 1 1 $\mathbb{P}^1 \subset \mathrm{Sym}^2 E$	1 2 2 $\mathbb{P}^2 \subset \mathrm{Sym}^3 E$	1 3 3 $\mathbb{P}^3 \subset \mathrm{Sym}^4 E$ 1
2	0 0	1 3 3 $\mathbb{P}^3$ 1	2 8 8 ramified over deg 6 hypersurface 2	15 15 gen finite not finite
3	0 0	2/1 6 7 Coble quartic 4	2/1 16 26 finite 54	2/1 30 63
4	0 0	2/1 9 15 96	2/1 24 80 ≤ 810	2/1 45 255
. . .		6336	≤ 8780	

$\Leftarrow$ proved to be regular map

$\deg \varphi$		$\underset{=}{(g-1)(r^2-1)}$ $\dim \mathcal{M}_C(r, \mathcal{O}_C)$ $\underset{=}{r^g - 1}$ $\dim \lvert r\Theta \rvert$
explicit description in terms of φ		$\deg \varphi \cdot \deg \mathrm{Im}\, \varphi$

2/1 : $\deg \varphi = \begin{cases} 2, & C \text{ hyperelliptic} \\ 1, & \text{otherwise} \end{cases}$